

Protecting Philadelphians Via Data Vigilantism

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1 Problem Statement

The City of Philadelphia has one of the highest crime rates and auto-fatalities rates among cities with over 1 million people in the US. Despite having one of the lowest car ownership demographics for a city of its size, Philadelphia continues to struggle with high rates of traffic accidents and crime. With this in mind, we pose the following topic question in analysing the datasets:

How can we identify high crime and crash areas in Philadelphia and optimise the allocation of police stations for effective law enforcement and public safety?

Hence, the goal of this study is to identify areas of high crime and crashes in Philadelphia and determine the optimal allocation of police stations to improve public safety. By analyzing crime and crash data, along with data on police stations and districts, we aim to provide novel insights and practical recommendations on how to reduce crime and improve road safety in the city.

2 Executive Summary

In this study, we conducted an analysis of crime and crash data in Philadelphia, along with data on police stations and districts, in order to identify areas of high crime and crashes and determine the optimal allocation of police stations. We utilised data cleaning techniques, exploratory data analysis, and unsupervised machine learning algorithms such as K-means clustering to identify hotspots of crime and crashes. Our results indicate that there are certain areas in Philadelphia with higher rates of crime and crashes, and that the allocation of police stations could be improved to better cover these high-impact areas.

3 Technical Exposition

3.1 Initial Analysis

In this study, we conducted an exploratory data analysis of crime rates, traffic accidents, and police activities in the Philadelphia area. Our aim was to identify areas of high crime and crashes and determine the optimal allocation of police stations to improve public safety in the region.

3.1.1 Traffic Stops

To provide insight into traffic stops in Philadelphia, we looked into the split between the two types: pedestrian vs vehicular, and plotted this on a scatter graph. Furthermore, a heat map was created to visualise the arrests made. The map was generated using the coordinates of the stops, and a HeatMap layer was added to represent the concentration of arrests made during the stops. The goal of this analysis was to identify any patterns or biases in the stops, which could help inform future policy decisions. By visualizing the data on a map, it was possible to identify areas with a higher concentration of stops and/or arrests.

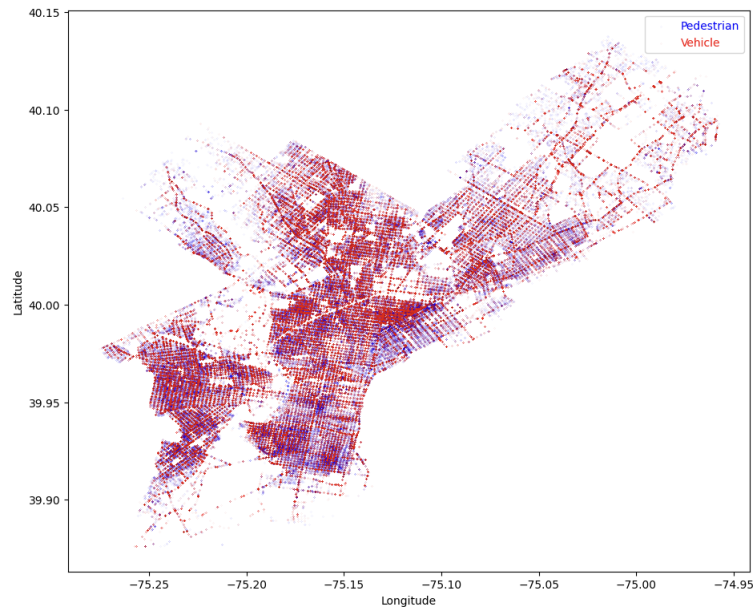


Figure 1: Traffic stops by type

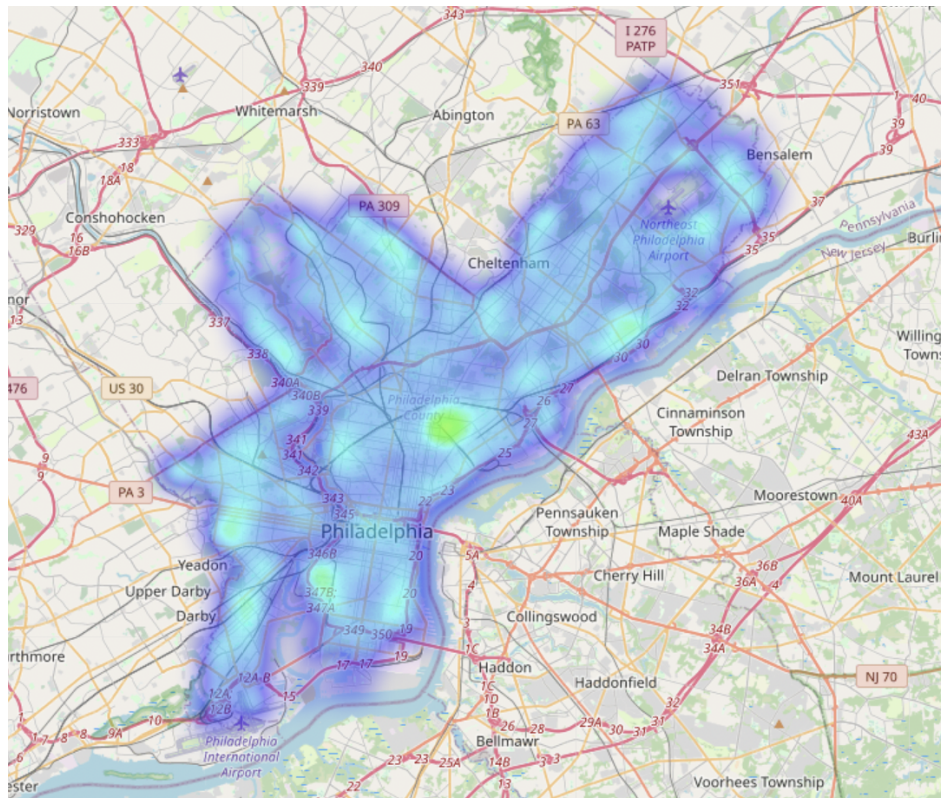


Figure 2: Location of arrests made at traffic stops

3.1.2 Crime Rates

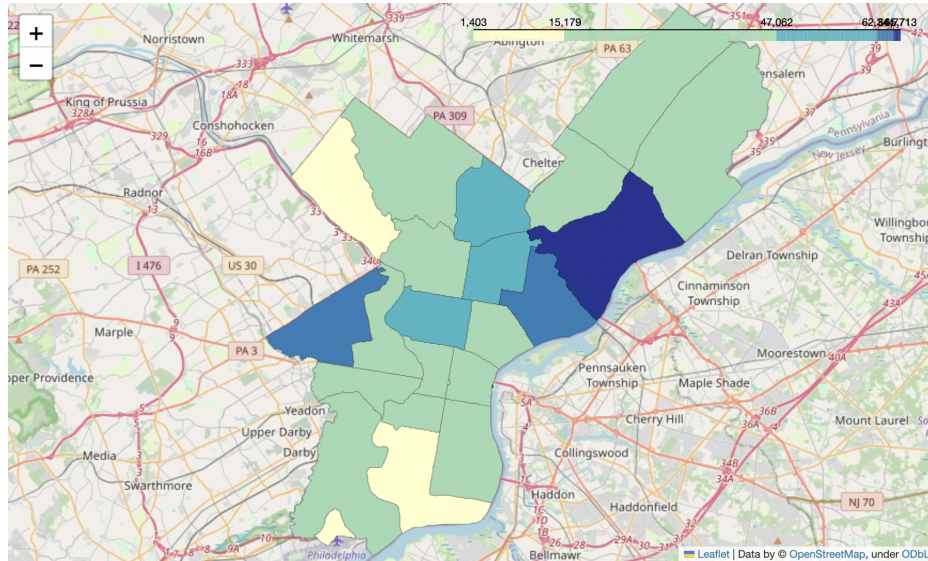


Figure 3: Crime rate density based on police district from 2014 - 2018

We examined the distribution of crime rates across police districts by plotting a choropleth map that showed the crime rate density for each district. Our analysis revealed that the highest concentration of crimes was in the central region of the city. This finding suggests that this area may require more attention from law enforcement to address the high incidence of criminal activity. Additional analysis could explore the potential reasons for this concentration of crime and develop targeted interventions to reduce it.

3.1.3 Time Required to Arrive at Traffic Accidents

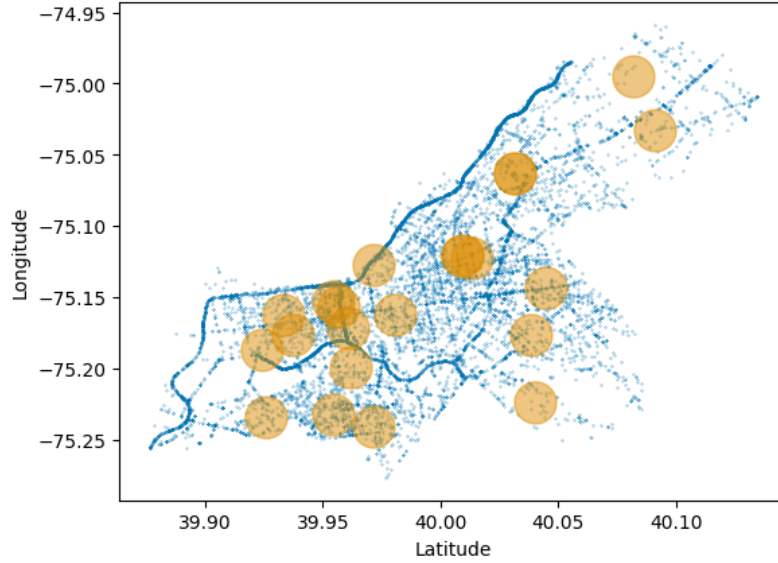


Figure 4: Map showing police stations in orange with radius 500m. The blue dots represent places that take more than 15 minutes for the police to arrive

We conducted an investigation into the relationship between the distance to the nearest police station and the response time of police to traffic accidents. While we initially anticipated that a closer distance would result in a faster response time, our analysis found no significant correlation between the two variables. However, we did identify a noteworthy trend: the majority of accidents with longer response times occurred on highways.

This finding suggests that factors beyond the proximity of police stations play a crucial role in emergency response times, such as the volume of traffic, road conditions, and accident severity. To optimize emergency response times and reduce the likelihood of fatalities and injuries in highway accidents, it may be necessary to examine these additional factors in more detail. Further research could explore potential strategies, such as improved road maintenance, traffic flow management, and communication protocols, to enhance the effectiveness of emergency response efforts on highways.

3.1.4 Police Investigations

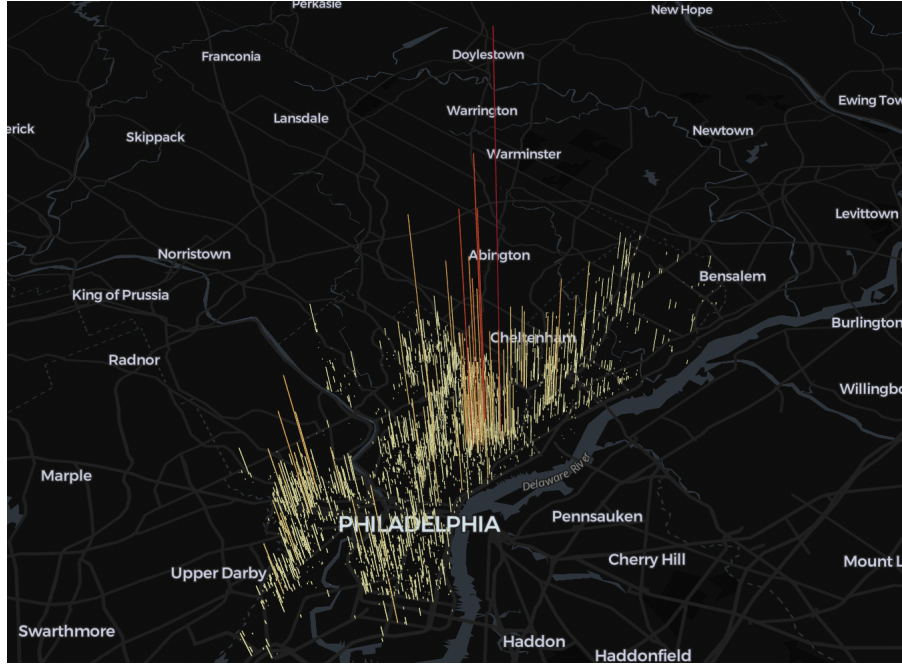


Figure 5: 3D plot showing the density of investigations

Furthermore, we conducted a density analysis on police investigations, which revealed a strong correlation between the density of investigations and the density of crime in the area. Notably, the city center emerged as the most densely investigated area, implying that it might be a hotspot for criminal activities. This finding emphasises the need for increased law enforcement resources to address crime effectively in this region. Additional analysis could explore the types of crimes that are most common in this area and develop targeted interventions to reduce their incidence.

3.1.5 Identifying Police Activity Patterns

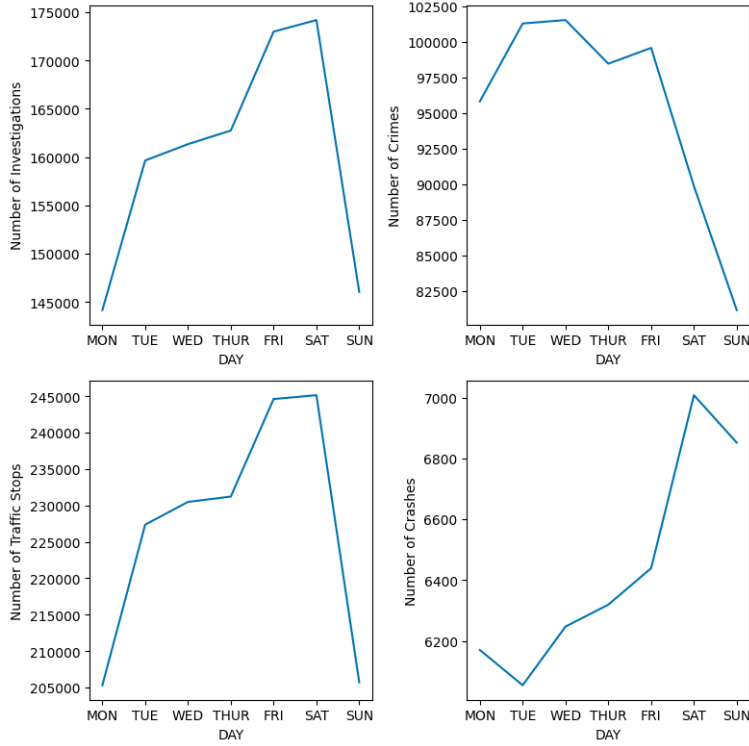


Figure 6: Investigations, Crimes, Traffic Stops and Crashes by Day

We also examined police activities, such as investigations and traffic stops, to identify patterns by day, month, and year. This analysis revealed seasonal and daily trends in police activities, with higher rates of traffic stops during the summer months and on weekends, and lower rates of investigations during the winter months.

These findings have practical implications for optimizing the allocation of police resources and enhancing the effectiveness of police activities in Philadelphia. For instance, our analysis can help the authorities to better allocate police officers to high-risk areas and times of the year, thereby improving public safety.

Overall, our exploratory data analysis provided valuable insights into the distribution of crime and crashes in the Philadelphia area, as well as the factors that influence police response times and patterns in police activities. By leveraging

these insights, law enforcement agencies can allocate resources more efficiently and effectively to enhance emergency response and maintain public safety in the region. Further analysis could explore the potential underlying reasons for the observed patterns and develop targeted interventions to address the root causes of crime and crashes in the area.

3.2 K-Means Clustering

3.2.1 Data Selection

To aid our initial interest in analysing crimes and crashes in Philadelphia via clustering, we focused on 6 datasets (see Table 3.2.1). To reduce the dimensionality of the data, we hand-picked features which captured either a crime or a crash. Hence, information pertaining to the time of day the event took place at or the weather at the time was deemed inconsequential and removed. A comprehensive list of features used can be found in the the Appendix (Table 2).

For the investigations dataset we selected the binary variables discerning whether an individual or a vehicle had been frisked or searched, if contraband had been found and if an arrest was made. The investigations dataset is one which could pertain bias and so may not exactly be a representative sample of actual crime. There is a host of literature relating to policing bias. Lum and Isaac (2016) showed that black people in Oakland were disproportionately arrested for drug use by comparing drug arrests with survey results about drug use. While illicit drug use existed everywhere, arrests tended to only occur in very specific locations, like low-income areas with minority populations. If this can happen with drug use then there is no reason to suspect that it doesn't happen with other types of crime and in other parts of America. Indeed Gelman et al. (2006) also found evidence of bias in an analysis of NYPD's Stop-and-Frisk Policy. They concluded that persons of African and Hispanic descent were stopped more frequently than Whites, even after controlling for precinct variability and race-specific estimates of crime participation. The key consequence here is that if we provide a machine learning algorithm with prejudiced data then it will learn these prejudices. The algorithm is doomed to replicate and amplify these biases. Worse still, this results in a vicious feedback loop as the more time police spend in one area the more crime they will eventually find. It's a case of selection bias meeting confirmation bias. We include this data, despite our aforementioned concerns, since arrests and contraband found may represent crimes being committed.

Dataset	Observations	Features	Descriptions
philadelphia_population_metrics	384	14	Population demographic information of the city of Philadelphia.
crash_info_general	133,013	101	Information related to car crashes in Philadelphia from 2010-2021.
crash_info_flag_variables	133,013	111	General information related to car crashes in Philadelphia from 2010-2021 but encoded with flag variables.
traffic_stops_philadelphia	1,865,096	23	A dataset of traffic stops in the city of Philadelphia from December 2013- April 2018.
investigations	2,785,087	21	A dataset of investigations taken by Philadelphia police between 2014-2022 on pedestrians and automobiles.
crimes	1,593,142	10	A dataset of crimes committed in Philadelphia from 2012-2022.

Table 1: Description of datasets used for K-Means clustering.

The crimes dataset contains a feature which contains unique text relating to the type of crime committed. We used one-hot encoding to create 33 binary features. The crimes documented were violent in nature, e.g. murder, armed robbery, rape etc., but there is an all other offences option which likely incorporates non-violent crimes. During our analysis we dropped the feature for gross negligence homicide as none were committed in the time period under observation.

We used both the general crash information dataset and the flag variables dataset, which contain the same number of observations. They can be joined via the crash record number (CRN). The difference between the datasets is that the flag variables dataset contains binary features relating to crashes. From this we used features which related to dangerous or illegal driving acts, e.g. whether the crash was alcohol related, whether a fatality occurred etc. For the purpose of this analysis we did not consider variables such as the number of bicycles involved or the number of commercial vehicles involved since this does not assist the allocation of policing resources.

In order to account for population density, since we assume a positive relationship between crimes or crashes with population, we included the number of people living in each census tract. It would be preferable to then compute statistics based on per capita, but 3 census districts contain a population of 0, so division is not possible. We could remove these areas from analysis and continue but this would represent an important data loss. For example, the area corresponding to GEOID 42101980600 has a population of 0 and contains the Wells Fargo Centre, multi-purpose indoor arena which is the home of the Philadelphia 76ers (NBA) and the Philadelphia Flyers (NHL). An area which has national exposure throughout the year is obviously important from a policing perspective. Between 2014 and 2017 this area had 863 crimes committed, 50 arrests made through investigations, 34 arrests made via traffic stops and 44 crashes involving injured parties.

External Data

The city of Philadelphia contains 22 police districts. One could analyse statistics based on each district but we can be even more precise and separate crimes and crashes by census tract. Using the census tract we have 384 areas as opposed to 22. The U.S. Census Bureau and other state and federal agencies use geographic identifiers, or GEOIDs, to divide areas for administrative and legal purposes. The GEOIDs can be up to 40 digits long, with each extra digit subdividing the areas further. The police districts are essentially an amalgamation of census tracts. Thus analysing with a more granular view allows us to identify areas within each district that are particularly high or low in crimes and crashes. We sourced a dataset from <https://opendataphilly.org>, corresponding to the census tracts which have been in place since 2010. This matches up with our earliest dataset, the population metrics for Philadelphia, and every other dataset

post-dates this. From the same website we also sourced the police district areas. Figure 7 shows the police districts overlaid on a map of the census tracts for Philadelphia. The 22 police stations are also shown. Note that despite there being an equal number of stations and districts, some districts have 2 stations while others have none.

Joining Data

To combine all the datasets we first needed to filter by a common time frame. The traffic stops dataset had the shortest date range, from the beginning of 2014 to the middle of 2018, and all other datasets overlapped this particular range.

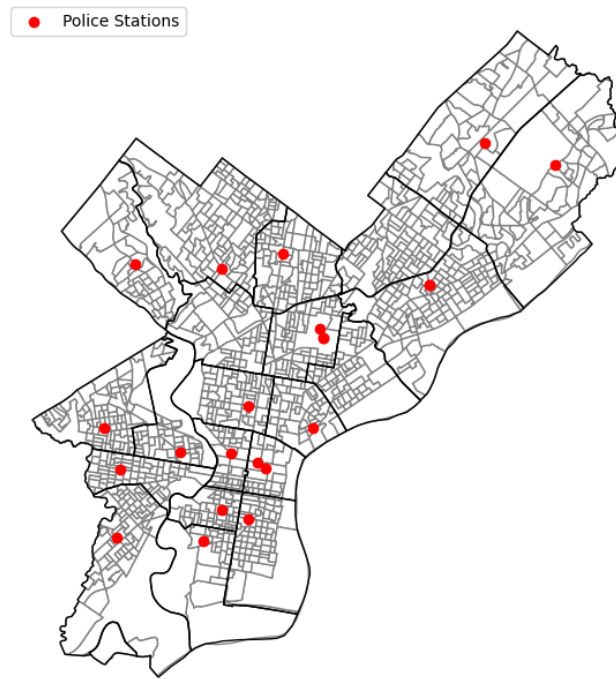


Figure 7: Map of Philadelphia with police districts outlined in black and census tract areas outlined in grey. Police stations are annotated by red markers.

We filtered the data between 2014 and 2017 to get 4 full years of information for all the datasets. We then grouped the data by census tract. Each observation in the datasets came with a 12 digit GEOID. This corresponds to the state, county, tract and block. While it is more precise than our 11 digit GEOID as it includes block, we do not have the required areas to incorporate blocks. Thus we grouped each observation by the first 11 digits. In grouping the data, we summarised the features by summing the values. Hence, each observation now corresponded to an 11 digit GEOID identifying a census tract, and each feature was the sum of events between the beginning of 2014 and the end of 2017.

Transforming the Data

Due to the skewed nature of the data, with most being right-skewed, we transformed the features by taking the square root. We avoided a log transformation as 0 was possible in our sample space. We then standardised the features for clustering.

3.2.2 Optimal K

To cluster each census tract based on their crime and crash information between the beginning of 2014 and the end of 2017, we first need to pre-assign the number of clusters to look for in the K-means algorithm. Selecting K is most often done using a plot of the within-cluster sum of squared (or inertia) against the number of clusters, K. The within-cluster sum of squares measures the dispersion of the observations in each cluster. A smaller value indicates that the clusters are densely compact. However, the within-cluster sum of squares monotonically decreases with K, reaching a minimum of 0 when the number of clusters equals the number of observations. Hence, identifying K is done by using the heuristic of looking for the elbow in the plot. We identified K=2 as the optimal number of clusters from Figure 8. This value of K produces the sharpest decrease in the within-cluster sum of squares.

As the elbow plot is not a bullet proof method of choosing K, we also analysed the silhouette plots for each K. A silhouette score ranges between -1 and 1. A score of -1 means the observation is better suited in another cluster, while a score of 1 means an observation is perfectly clustered. Values around 0 indicate that an observation is on the boundary of two clusters. Figure 9 shows the silhouette scores for each observation as well as the proportion of observations in each cluster. Most observations have been clustered appropriately, bar 31 observations with a negative silhouette score. We then produced the same plots for K=3 in Figure 10. We see cluster 0 and cluster 1 when K=2 equate to cluster 2 and cluster 1 when K=3, respectively. The cluster 0 for K=3 actually contains mainly values from cluster 1 when K=2. This indicates to us that K=2 is preferable to K=3. There are also an equal number of observations with

negative silhouette score (31) and the average silhouette score has decreased from 0.3 to 0.25.

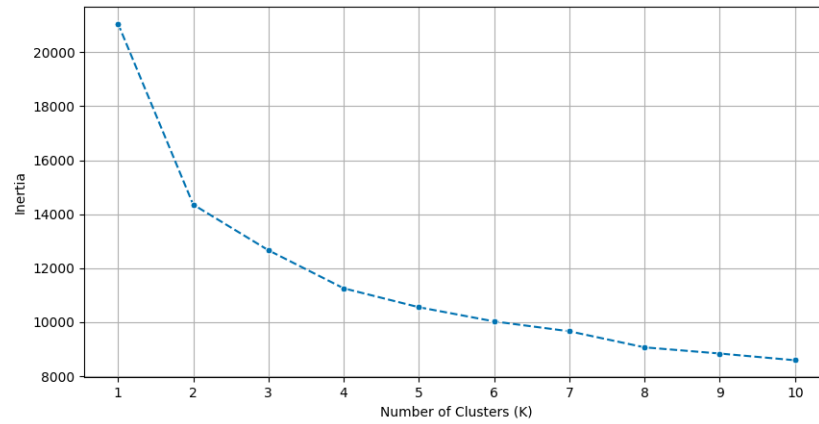


Figure 8: Plot of the within-cluster sum of squares(inertia) against the number of clusters K.

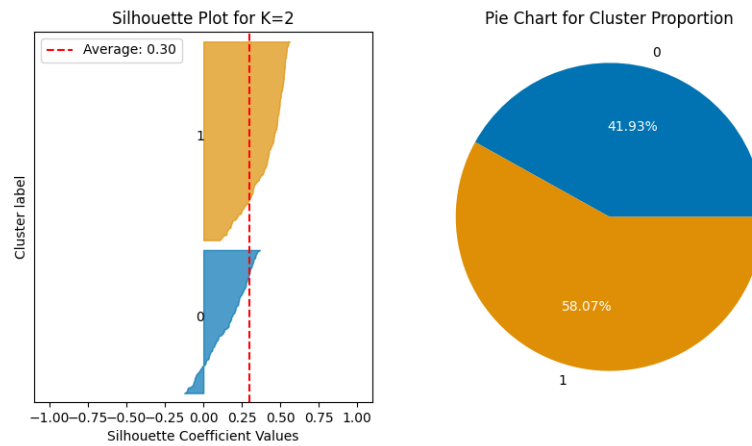


Figure 9: Silhouette plot and pie chart showing proportion of observations for K=2.

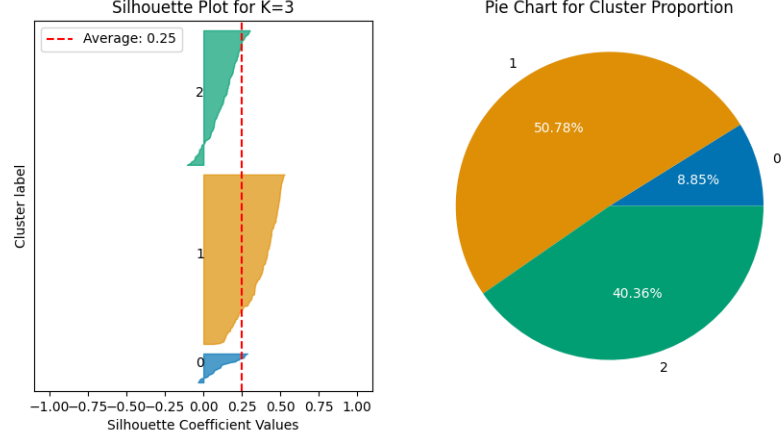


Figure 10: Silhouette plot and pie chart showing proportion of observations for $K=3$.

3.2.3 Clustering Results

A table of the cluster means can be found in the Appendix (Table 3). Cluster 0 identified areas that are generally higher in population, dangerous driving, arrests made and violent crimes. Figure 11 shows the distribution of the population for both clusters, Figure 12 shows the same plots for arrests made at traffic stops while Figure 13 shows these plots for rapes in each cluster. We associate cluster 0 with areas requiring more policing than in cluster 1. However, this is not a call to provide the area with more policing. In fact frisks were carried out 10 times more in areas in cluster 0 than areas in cluster 1, despite the mean population being only 1.3 times the other. Frisking and searching may yet be justified as the number of arrests are 9 times that of cluster 1. The ratio of frisks to arrests is higher in cluster 0 than in cluster 1. With more time a review of bias in these areas could've been carried out, with some statistical tests to distinguish whether in fact there is a difference.

The biggest discrepancy is between gambling violations, with 36 times more arrests made in cluster 0 for this (Note rounding in the table makes it appear 24 but it is actually 36). Gambling violations are relatively rare however, so this is affected by small sample size. Areas in cluster 0 are also far more likely to have prostitution and commercialized vice, as well as homicides of all kinds.

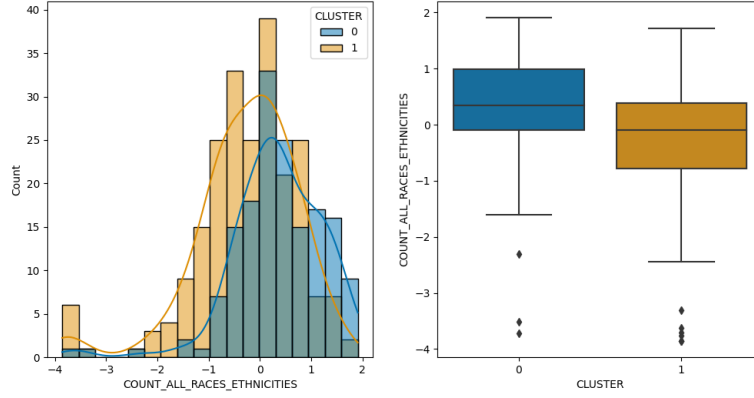


Figure 11: Histogram and boxplot of population for each cluster.

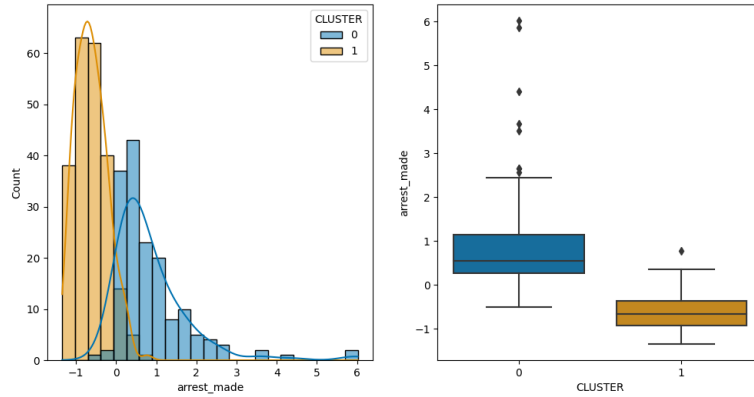


Figure 12: Histogram and boxplot of arrests made at traffic stops for each cluster.

3.2.4 Comparing Clusters with Actual Districts

With the funding of policing in the U.S. becoming a politicised and prevalent issue we believe it is worth comparing our clusters with the actual police districts and the location of the police stations. Figure 14 shows the map of Philadelphia with an outline of police districts and census tracts and police station locations highlighted similar to the plot shown earlier. However, we have coloured each

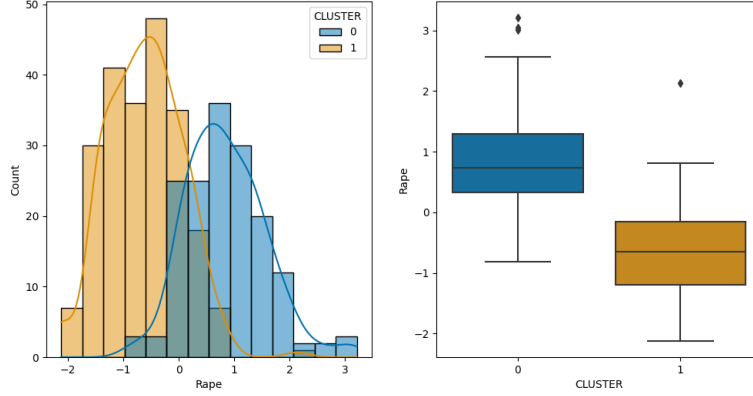


Figure 13: Histogram and boxplot of rapes for each cluster.

tract based on the cluster they have been assigned to. Areas in cluster 0, which we associated with higher incidences of crimes and crashes, have been coloured blue, while areas in cluster 1 are coloured in yellow.

This maps highlights a number of things. Firstly, police stations for district 39 and 24, which are located close to each other, are in cluster 0. Our findings would suggest that this is likely a suitable place for these two stations as the areas have higher populations densities and crime rates than most. However, police districts 6 and 77 (which is labeled as 0 in the figure) are likely not in optimal locations.

3.3 Conclusion and Discussions

In conclusion, our analysis using unsupervised learning has highlighted the need for a reevaluation of the placement of police stations and their respective districts. Our clustering model revealed that multiple police stations were grouped together in one cluster, while some high-risk clusters had no nearby police stations.

This suggests a potential issue with the current distribution of police resources in the area. By adjusting the location of police stations or increasing the number of patrols in high-risk areas, law enforcement could potentially improve response times and reduce crime rates in these areas.

We recommend that the Philadelphia government consider reallocating po-

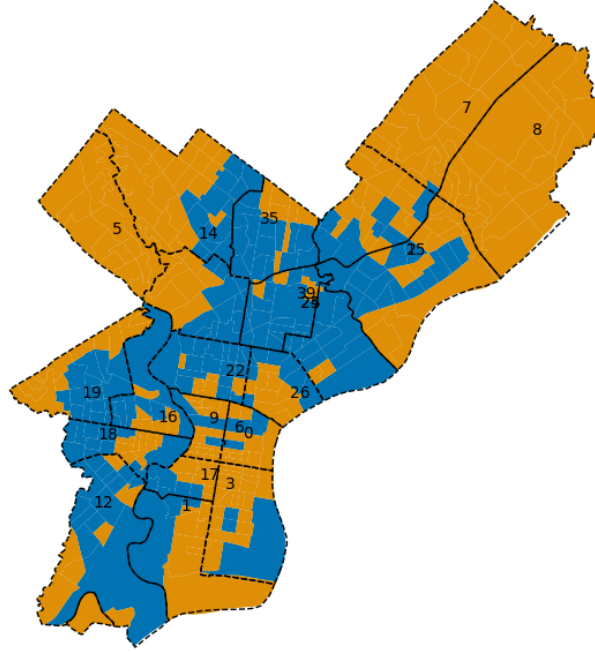


Figure 14: Map of Philadelphia with police districts outlined in black and census tracts outlined in white. Blue areas are cluster 0 and yellow areas are cluster 1. Police stations are annotated with their district number. Note district 0 is actually district 77.

lice stations based on the clusters identified in our analysis. Specifically, allocating more police stations to higher-risk clusters could help to improve safety and reduce crime in those areas.

However, it is important to note that our analysis only took into account the geographical distribution of police stations and crime incidents. Future research could consider additional factors such as socioeconomic status and historical crime patterns to create a more comprehensive understanding of the optimal placement of police resources.

Additionally, our analysis was limited by the time available for this study. We relied on reported crime incidents and police station locations, which may not fully reflect the entire scope of criminal activity in the area. Future studies could incorporate data from other sources such as emergency calls, citizen reports, and social media to gain a more complete picture of crime patterns in the area.

Overall, our analysis provides valuable insights into the distribution of police resources in Philadelphia and highlights the potential benefits of using unsupervised learning techniques to inform resource allocation decisions. By prioritising the allocation of police resources to high-risk areas, law enforcement can work towards creating a safer and more secure community for all residents.

References

- A. Gelman, A. Kiss, and J. Fagan. An analysis of the nypd's stop-and-frisk policy in the context of claims of racial bias. *Columbia Public Law Research Paper*, (05-95), 2006.
- K. Lum and W. Isaac. To predict and serve? *Significance*, 13(5):14–19, 2016.

A Appendix

Dataset	Feature
investigation	individual_frisked individual_searched individual_arrested individual_contraband vehicle_frisked vehicle_searched vehicle_contraband
crimes	Aggravated Assault Firearm Aggravated Assault No Firearm All Other Offenses Arson Burglary Non-Residential Burglary Residential DRIVING UNDER THE INFLUENCE Disorderly Conduct Embezzlement Forgery and Counterfeiting Fraud Gambling Violations Homicide - Criminal Homicide - Criminal Homicide - Gross Negligence Homicide - Justifiable Liquor Law Violations Motor Vehicle Theft Narcotic / Drug Law Violations Offenses Against Family and Children Other Assaults Other Sex Offenses (Not Commercialized) Prostitution and Commercialized Vice Public Drunkenness Rape Receiving Stolen Property Robbery Firearm Robbery No Firearm Theft from Vehicle Thefts Vagrancy/Loitering Vandalism/Criminal Mischief Weapon Violations
crash_info_flag_variables	AGGRESSIVE_DRIVING

	ALCOHOL_RELATED CELL_PHONE DRUG_RELATED FATAL INJURY PROPERTY_DAMAGE_ONLY SPEEDING_RELATED
traffic_stops_philadelphia	arrest_made frisk_performed search_conducted search_person search_vehicle raw_individual_contraband raw_vehicle_contraband

Table 2: Features used in clustering.

Feature	Cluster 0 mean	Cluster 1 mean
COUNT_ALL_RACES_ETHNICITIES	4428.61	3253.73
AGGRESSIVE_DRIVING	60.89	34.55
ALCOHOL_RELATED	6.94	3.31
CELL_PHONE	0.41	0.18
DRUG_RELATED	3.83	1.22
FATAL	0.63	0.23
INJURY	103.46	52.44
PROPERTY_DAMAGE_ONLY	22.96	17.15
SPEEDING_RELATED	11.13	6.99
arrest_made	368.07	56.41
frisk_performed	682.14	86.30
search_conducted	440.86	65.33
search_person	368.48	57.23
search_vehicle	127.54	12.90
raw_individual_contraband	116.26	15.21
raw_vehicle_contraband	43.05	4.91
individual_frisked	434.61	40.99
individual_searched	259.33	29.44
individual_arrested	260.71	28.84
individual_contraband	81.71	7.40
vehicle_frisked	129.48	8.76
vehicle_searched	95.15	6.89
vehicle_contraband	31.23	2.45
Aggravated Assault Firearm	40.48	5.45
Aggravated Assault No Firearm	91.48	23.75
All Other Offenses	576.30	122.68

Arson	6.01	1.31
Burglary Non-Residential	17.07	8.97
Burglary Residential	86.00	42.41
DRIVING UNDER THE INFLUENCE	59.30	16.49
Disorderly Conduct	26.34	5.37
Embezzlement	2.45	2.07
Forgery and Counterfeiting	2.32	1.00
Fraud	156.84	78.24
Gambling Violations	0.24	0.01
Homicide - Criminal	2.92	0.14
Homicide - Criminal	0.29	0.03
Homicide - Justifiable	0.00	0.00
Liquor Law Violations	0.73	0.39
Motor Vehicle Theft	25.43	10.42
Narcotic / Drug Law Violations	142.13	14.63
Offenses Against Family and Children	1.96	0.14
Other Assaults	373.03	128.17
Other Sex Offenses (Not Commercialized)	14.47	5.55
Prostitution and Commercialized Vice	7.40	0.25
Public Drunkenness	3.04	0.50
Rape	20.49	5.28
Receiving Stolen Property	0.37	0.02
Robbery Firearm	46.12	11.67
Robbery No Firearm	57.39	17.69
Theft from Vehicle	160.11	104.14
Thefts	274.86	146.35
Vagrancy/Loitering	1.06	0.10
Vandalism/Criminal Mischief	231.45	94.01
Weapon Violations	31.09	4.03

Table 3: Cluster means for K-Means clustering.